

Drug-loaded mesoporous silica on carboxymethyl cellulose hydrogel: Development of innovative 3D printed hydrophilic films

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ABSTRACT

3D printing has been explored as an emerging technology for the development of versatile and printable materials for drug delivery. However, the alliance of 3D printing and nanomaterials has, to date, been little explored in pharmaceutics. Herein, a mesoporous silica with nanostructured pores, SBA-15, was used as a drug carrier for triamcinolone acetonide, a hydrophobic drug, with the aim of incorporating the drug formulation in a hydrophilic printable ink. The adsorption of the drug in the SBA-15 pores was confirmed by the decrease in its surface area and pore volume, along with an increase in the apparent aqueous solubility of triamcinolone acetonide, as shown by *in vitro* release studies. Thereafter, a hydrophilic ink composed of carboxymethyl cellulose containing drug-loaded SBA-15 was formulated and 3D printed as hydrophilic polymeric film using the semisolid extrusion technique (SSE). The 3D printed films showed complete drug release after 12 h, and the presence of the triamcinolone acetonide-loaded SBA-15 improved their *in vitro* mucoadhesion, suggesting their promising application in oral mucosa treatments. Besides representing an innovative platform to develop water-based mucoadhesive formulations containing a hydrophobic drug, this is the first report proposing the development of SSE 3D printed nanomedicines containing drug-loaded mesoporous silica.

1. Introduction

Mesoporous silicas are inorganic structures with nanopores of uniform size, which are arranged in cubic, hexagonal or lamellar arrangements. The arrangement and size of the pores is dependent on a structure-directing agent, which acts as a template during the synthesis of the silica (Arriagada et al., 2019). SBA-15 type mesoporous silicas were first reported in 1998, with a hexagonal pore arrangement and pore diameter close to 6 nm (Zhao et al., 1998). In addition to the narrow pore distribution, they have a large surface area and large pore volume, which provides large reservoirs for drug encapsulation. Due to this high surface area, SBA-15 increases the dissolution rate and the apparent solubility of drugs that are poorly soluble in water. Therefore, this encapsulation has been a good alternative to improve the

performance of lipophilic substances in topical application, promoting a homogeneous release of the drug (Jadhav et al., 2017; Montazeri et al., 2020; Narayan et al., 2018).

The development of pharmaceutical formulations with dispersed silica-based nanostructures has presented an interesting alternative for topical treatments. Authors have demonstrated a sustained release of nonivamide incorporated into porous silica for the treatment of chronic pruritus (Heck et al., 2016), and the SmartPearls technology was used to stabilize the amorphous state of poorly soluble drugs in mesoporous silica particles (Müller et al., 2019).

In parallel, the alliance between nanotechnology and 3D printing has emerged as an interesting and versatile approach in the development of innovative pharmaceutical formulations (de Oliveira et al., 2021; dos Santos et al., 2021). The role of some nanomaterials, such as

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nanocapsules, cyclodextrins, nanocrystals and inorganic nanoparticles, has been reported to improve the physical properties of printable inks (Wu, 2020; Yoon et al., 2021), as well as a strategy to tailor the biopharmaceutical properties of pharmaceuticals (Beck et al., 2017; Germini and Peltonen, 2021; Katsiotis et al., 2021). MCM-41 mesoporous silica was recently used to overcome the challenges in the oral administration of 3D printed drug dosage forms with poor soluble drugs. Up to now two printing techniques have been used this approach: fused deposition modelling (FDM), a hot extrusion printing technique, with polyvinyl alcohol as polymer (Katsiotis et al., 2021), and inkjet printing, with a pharmaceutical nano-ink suspension printed onto hydroxypropylmethyl cellulose and polyester transparency substrates (Wickstrom et al., 2017).

Among the available 3D printing techniques, semisolid extrusion (SSE) has become the most used in recent years for the development of biomedical and pharmaceutical products, overcoming FDM. This rapid growth can be explained by the variety of hydrogels available in tissue regeneration and the easy modulation of semisolid properties to attain good printability, as well as the easy control of the printing parameters (dos Santos et al., 2021). In the development of drug dosage forms, SSE presents the benefit of printing at room temperature, which is crucial for thermo-labile drugs (Kim et al., 2022). The selected polymer for printing must also provide good extrudability and printability, which can be usually linked to the rheological properties of inks. Hydrophilic polymers usually match these conditions, but appropriate dispersion of the drug in the polymer matrix must be studied and ensured (Kim et al., 2022; Panraksa et al., 2021), which sometimes can be only reached using organic cosolvents and high percentage of surfactants.

The alliance of these two emergent technologies, nanotechnology and 3D printing, might be interesting in the development of personalized hydrophilic films containing drugs with low aqueous solubility for oral mucosa application (Karavasili et al., 2021) without using organic solvents or surfactants. In recent decades, much has been studied on the use of the oral mucosa as a route for drug administration. The use of synthetic glucocorticoids, such as triamcinolone acetonide (TA), is widely prescribed for the treatment of oral lesions. However, the available pharmaceutical forms require frequent application, as the region undergoes mechanical action due to swallowing, salivation and the presence of mucus and enzymes, which makes it difficult to retain the drug at the site of action (Amiri Dehkharghani et al., 2018; Caro et al., 2017; Shin and Kim, 2000). Formulations containing high portions of liquid paraffin have been used due to their mucoadhesive properties, which prolong the drug residence time at the site of action.

Given the above, we hypothesized that loading a hydrophobic drug, such as TA, on mesoporous silica could improve its solubility, making it feasible to disperse it in hydrogels with 3D printable properties. To the best of our knowledge, there are no previous reports on the development of 3D printed dosage forms containing drug-loaded silica nanomaterials using the SSE technique. Therefore, the aim of this study was to develop a 3D-printed hydrophilic and mucoadhesive polymeric film containing a hydrophobic drug (TA) loaded in mesoporous silica SBA-15.

2. Materials and methods

2.1. Materials

Triamcinolone acetonide (TA) was obtained from Valdequímica (São Paulo, Brazil), carboxymethyl cellulose (CMC) 4000 was obtained from Leviale (São Paulo, Brazil), tetraethyl orthosilicate 98% (TEOS) and Pluronic P123 were purchased from Sigma-Aldrich, ethanol from Química Moderna, and HCl 37% and acetonitrile from Merck. All other reagents and solvents were of analytical or pharmaceutical grade.

2.2. Synthesis of SBA-15 mesoporous silica

For the synthesis of SBA-15 particles, 4 g Pluronic P123 was added to

a round-bottom flask containing 150 mL of 1.6 mol L⁻¹ HCl aqueous solution. The mixture was magnetically stirred at 40 °C until the complete dispersion of the surfactant. Then, 9.1 mL of TEOS were slowly dripped in to the dispersion, and a white precipitate appeared soon after. After stirring for 24 h at 40 °C, the mixture was transferred to a stainless steel autoclave with an internal Teflon cup, and the system was sealed and heated up to 100 °C for 24 h. Later, the precipitate was filtered and washed thoroughly with distilled water, in order to remove excess Pluronic P123. The obtained solid was dried in an oven at 80 °C, and calcination was then carried out at 550 °C for 6 h for complete surfactant removal.

2.3. TA loading on SBA-15 mesoporous silica

TA was incorporated into the SBA-15 by the incipient wetness method (Li et al., 2019). For this, an ethanolic solution of the drug was prepared at a concentration of 10 mg mL⁻¹. Then, 1428 µL of this solution was added to 200 mg of previously dried SBA-15, in order to obtain the dose of 1 mg of TA encapsulated in 15 mg of formulation (6.67% w/w). The solid was kept at 60 °C for 24 h for solvent evaporation and is hereafter referred to as SBA-TA.

The same method was used to prepare the blank formulation, without drug, called SBA-B, where 1428 µL of ethanol was added to the same amount of dry SBA-15 (200 mg).

2.4. Characterization of drug-loaded mesoporous silica

2.4.1. Physicochemical and morphological characteristics

The textural characterization of the SBA-15 mesoporous silica, before and after TA incorporation, was carried out through N₂ sorption analysis (Tristar II Kr 3020, Micromeritics, Georgia, USA). The samples (n = 3) were degassed beforehand, at 60 °C for 12 h. The specific surface area was determined by the BET (Brunauer, Emmett and Teller) multi-point technique, and the pore size distribution was obtained using the BJH (Barret, Joyner and Halenda) method (Gregg and Sing, 1982; Webb et al., 1997).

The amount of TA incorporated in the SBA-15, as well as the decomposition profile of the TA were evaluated by thermogravimetric analysis (TGA), using the Shimadzu Instrument equipment model TGA-50 (Kyoto, Japan). The samples were heated from room temperature to 900 °C, at a rate of 10 °C min⁻¹, under a N₂ flow of 50 mL min⁻¹.

The morphological characterization of the SBA-15 particles was performed by scanning electron microscopy (SEM) with a Field Emission Gun microscope (Inspect F50, FEI, Oregon, USA). The samples were prepared by dispersing the powder on a carbon conductive tape adhered to the aluminium support, which was then covered with a thin film of gold. Transmission electron microscopy (TEM) was performed to evaluate the SBA-15 mesostructure in a JEOL JEM-1011 microscope (Peabody, USA) at 100 kV acceleration voltage, using a carbon-coated Cu grid.

Differential scanning calorimetry (DSC) analyses were performed using a differential scanning calorimeter (Shimadzu DSC-60, Kyoto, Japan), at a heating rate of 10 °C min⁻¹, from room temperature to 330 °C. N₂ gas was used at a flow rate of 50 mL min⁻¹. The following samples were analysed: SBA-15, crystalline TA (raw material), physical mixture (SBA-15 + TA) and SBA-TA.

The hexagonal mesopore arrangement of the SBA-15 was assessed by X-ray powder diffraction, using the Siemens model D500 diffractometer, with CuKα radiation (λ = 0.15405 nm), a voltage of 40 kV and a current of 17.5 mA. The scanning range was 2θ from 0.2 to 6°. The crystallinity of TA when incorporated in the SBA-15 was also evaluated by extending the scanning range of 2θ up to 60° in the analysis of the following samples: SBA-15, SBA-TA and crystalline TA.

2.4.2. Drug content

The determination of TA content (n = 3) was assayed by high

performance liquid chromatography (HPLC) according to a method previously reported by Sallam et al. (2016). The system consisted of a Shimadzu LC-System (Kyoto, Japan) equipped with a LC-20AT pump, SPD-20AV detector (UV detector), DGU-20A5 degasser, CBM-20A system controller, SIL-20A auto sampler and LC-Solution, and a column reverse phase (250 × 4.6 mm ID, 5 µm particle size). The mobile phase was composed of acetonitrile-water (50:50 v/v), an isocratic flow rate of 1 mL min⁻¹, an injection volume of 20 µL and wavelength detection at 240 nm. The method was previously validated, considering the parameters of linearity, specificity, accuracy and precision.

To extract TA from the samples, the SBA-TA was homogenized and dried for 3 h at 60 °C to remove any residual moisture. Then, 19.8 mg of the sample was suspended in 10 mL of ethanol. The system remained under magnetic stirring at room temperature for 2 h, and the sample was then centrifuged at 10,000 rpm for 15 min. The supernatant was collected, diluted, filtered and injected into the chromatographic system.

2.4.3. *In vitro* release profile

The drug release studies of SBA-TA, an ethanolic solution of TA and a suspension of TA were performed by the dialysis bag technique, using a mixture of phosphate buffer solution pH 7.4:ethanol (70:30 v/v) as the medium. The TA saturation concentration in the release medium was established as 507 µg mL⁻¹, and sink conditions were maintained during the experiments. For the study with the ethanolic solution of TA, the drug was solubilized in ethanol at a concentration of 200 µg mL⁻¹, and 1.0 mL of the solution was added to the dialysis bag. For the analysis of TA in suspension, the same amount of drug and 1.0 mL of release medium were added to the dialysis bag. Likewise, SBA-TA samples equivalent to 200 µg of TA were weighed and placed in the dialysis bag with 1.0 mL of the medium. The dialysis bags were then inserted into 150 mL of medium and the systems kept under constant magnetic stirring at 37 °C. Aliquots of 1.0 mL were collected at predetermined times (5, 10, 15, 30, 60, 120, 180, 300, 480, 720, 1440 and 2880 min) and immediately replaced with fresh medium. Quantitative analyses were performed using the HPLC previously described.

2.5. Preparation of hydrogels and 3D printed films

The hydrogels were produced with CMC (5%), glycerin (10%), SBA-TA (0.7%) and water (84%), which were manually mixed with a porcelain mortar and pestle. The SBA-TA and the CMC were homogenized and dispersed in the glycerin, and the water was then incorporated slowly until a homogeneous hydrogel was formed (HG-SBA-TA). A plain hydrogel (HG) was prepared without SBA-TA for comparison. Once the semisolids were prepared, they were immediately packed into syringes. The attached piston was pushed to eliminate air in contact with the formulation.

The 3D printing was carried out by the semisolid extrusion technique using 3DBioEdPrinterV2 (BioEdTech, São Paulo, Brazil), with the following parameters and conditions: extrusion speed (5 mm s⁻¹), extrusion temperature (room temperature), filling percentage (50%), layer height (0.2 mm), nozzle diameter (0.25 mm) and dimensions (8x8x2 mm). The printing time for each film was 19 min.

2.6. Characterization of hydrogels and 3D printed films

2.6.1. pH of hydrogels

Samples for pH measurements were prepared by adding 0.5 g of the hydrogels (HG and HG-SBA-TA) to 5 mL of ultrapure water and stirring for 1 h. After complete dispersion, the pH was measured at room temperature, in triplicate.

2.6.2. Rheological measurements of hydrogels

Rheological properties were evaluated using an Ares G2 rheometer (TA instruments, New Castle, DE, USA) with a 25 mm parallel plate

geometry set at a 1 mm gap and at 20 °C. A flow ramp test was carried with a shear rate range of 0.8 to 4.0 1/sec to determine the best fit flow for the samples. Yield stress value (Pa) and the linear viscoelastic region (LVR) were estimated through an oscillatory amplitude test with a stress sweep of 0.1–500% at a constant frequency of 1 Hz. Viscoelastic properties (storage modulus (G'), loss modulus (G'') and tangent delta (tan δ)) were evaluated at a constant strain of 0.1% and angular frequency sweep of 0.2 to 20 rad/s. All measurements were carried out in triplicate.

2.6.3. Mean weight, size and thickness of films

The film weight (n = 10) was measured immediately after printing (while still wet) and after drying for 48 h at room temperature. The dimensions and thickness of the films were measured after 48 h using a caliper.

2.6.4. Drug content of films

The drug content of the 3D printed films was assayed by HPLC. Films (n = 3) were individually placed in volumetric flasks containing a mixture of methanol:water (1:1) and sonicated for 2 h. The dispersion was centrifuged, and 1.0 mL of the supernatant was collected and transferred to a 5.0 mL volumetric flask with acetonitrile, remaining under magnetic stirring for another 2 h. The samples were then filtered and analysed by HPLC.

2.6.5. *In vitro* release profile from films

The *in vitro* drug release study from the 3D printed films was performed according to the method described in Section 2.4.3, where the film was placed in the dialysis bag with 2.5 mL of release medium and the bags were inserted in 100 mL of release medium under stirring at constant temperature (37 °C). Likewise, the system was maintained for 48 h, with 1.0 mL of medium being collected at predetermined periods.

2.6.6. Swelling index of films

Swelling indexes of films with and without SBA-TA were determined by weighting each film and placing them, individually (n = 4), in a Petri dish glass containing 7 mL of ultrapure water. In predetermined times (1, 2, 3, 4, 5, 7, 10, 12, 15, 20, 25, 30 min), the films were carefully removed from the dish, dried with soft paper and re-weighted. The experiment was finished when it was not possible to remove the film from the dish glass. At the end, the swelling index was calculated according to the equation (1).

$$\text{Swelling index (\%)} = \frac{(W_f - W_i)}{W_i} \times 100 \quad (1)$$

Where W_f is the final mass of hydrated films and W_i is the initial mass.

2.6.7. Disintegration time of films

The disintegration of the films (n = 3) was evaluated with a disintegrator (Nova Etica, model 301AC), with 700 mL of water as an immersion vehicle at 37 °C ± 0.5. The disintegration of 3D printed films from hydrogels with and without SBA-TA was analysed. The samples were added to the baskets and the immersion and removal movement was carried out in the vehicle until complete disintegration of the films.

2.6.8. Mechanical properties of films

Tensile strength of 3D printed films (n = 4), with and without SBA-TA, was measured with TA.XT plus Texture Analyser (Stable Micro Systems, Godalming, UK) equipped with a 50 kg load cell and tensile grips (A\TG). Films were printed in a special size for the test (50x8x2 mm) and placed vertically in the screw with a gauge length of 30 mm. Afterwards, each film was stretched at an elongation speed of 0.5 mm s⁻¹ until it was broken. Exponent Software (Stable Micro Systems, Godalming, UK) was used to record the values of force, distance, stress, and strain.

2.6.9. Mucoadhesion of films

Mucin discs were prepared according to a procedure previously reported (Chaves et al., 2018), using a TA.XT plus Texture Analyser (Stable Microsystem, Godalming, UK). The discs were produced by compression, using a test speed of 1 mm s^{-1} , a post-test speed of 1 mm s^{-1} and a distance of 4 mm. Before the mucoadhesion test, the discs were hydrated with 20 μL of ultrapure water, and excess water was removed with absorbent paper.

Mucoadhesion ($n = 3$) was evaluated in terms of the force needed to remove the mucin disc from the formulation as a function of time, using the same equipment. The mucin disc was fixed in the probe and the samples (SBA-15 and SBA-TA powders, plain CMC film and CMC film with SBA-TA) were fixed on the equipment's platform using an instant adhesive. A force of 290 mN was applied to maintain the contact of the formulations with the mucin for 180 s, being retracted at a speed of 0.10 mm s^{-1} (Chaves et al., 2018).

3. Results and discussion

Mesoporous silicas such as SBA-15 are known for their capacity for hosting high drug loads in their nanopores and increasing the drug's apparent solubility by dispersing it through its large surface area. These features, along with the intrinsic physicochemical characteristics of silica, support our hypothesis about the use of SBA-15 as a drug carrier in the development of water-based formulations of lipophilic drugs. Herein, SBA-15 particles were synthesized and used in the nano-encapsulation of TA, a poorly water-soluble drug. The loaded particles were further incorporated in a CMC hydrogel, aiming to develop a 3D printed oral mucosal drug delivery system.

3.1. TA-loaded on SBA-15 mesoporous silica (SBA-TA)

Morphological characterization of SBA-15 particles was performed by SEM, and Fig. 1A presents the acquired image. As can be seen, the particles are rod-like in shape and submicrometric in size, with dimensions ranging from c.a. 0.5 to $1.1 \mu\text{m}$. Even when aggregated in a rope-like structure, its size is still far below the macroscopic dimensions required for nozzle extrusion during a printing process. An elongated particle shape is reported for SBA-15 particles (Oliveira et al., 2020; Patel et al., 2021), which is related to the synthesis conditions, and mainly the pH, which modulate the condensation reactions of the silica precursor (Patel et al., 2021).

The SBA-15 pore structure cannot be assessed by SEM, due to the limitation of the technique; therefore, TEM analysis was performed. The typical cylindrical nanochannels of SBA-15, parallel and equidistantly arranged, can be observed in Fig. 1B, which shows the long-range ordering of the material (Oliveira et al., 2020). XRD analysis was also

used to evaluate the pore arrangement. The obtained diffractograms in the range of 2θ from 0.2 to 5° are presented in Fig. S1. The hexagonal array of uniform cylindrical nanochannels was evidenced by the characteristic diffraction pattern $2\theta = 0.9^\circ$, 1.5° and 1.75° , attributed to the reflection planes (100), (110) and (200) (Zhao et al., 1998). The loading of TA did not produce significant changes in the peak positions or peak relative intensities; therefore, the mesostructure of SBA-15 is preserved in the SBA-TA formulation.

The loading of TA in the SBA-15 nanochannels was performed by the incipient wetness method, and HPLC analysis was used to assess the experimental amount of drug incorporated into the SBA-TA formulation. This method was validated and found to be linear ($r = 0.9993$) in the range of 2.5 to $50 \mu\text{g mL}^{-1}$. The specificity, precision and accuracy of the method were also evaluated. The detection and quantification limits of the method were calculated as 0.25 and $0.77 \mu\text{g mL}^{-1}$, respectively. The estimated drug loading was $0.067 \pm 0.006 \text{ mg TA}$ for each 1 mg of SBA-TA, representing a drug loading of 6.7% . The experimental drug loading calculated by the HPLC drug assay was in agreement with the theoretical drug loading (6.67%), suggesting that there is no significant drug loss in the process.

The drug loading was also evaluated by TGA, along with the thermal profile of SBA-15 and SBA-TA samples. The obtained thermograms are presented in Fig. 2.

From the thermogram in Fig. 2, two regions of mass loss were observed. The first one, up to 150°C , was observed in both samples and is attributed to water desorption. The second region was observed between 150 and 900°C , where dehydroxylation reactions of silica occur

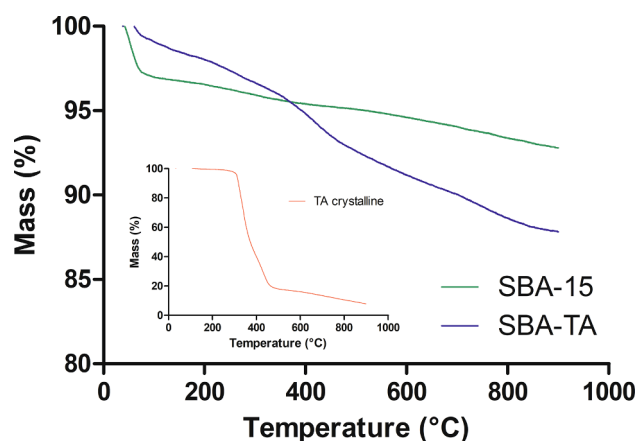


Fig. 2. Thermogravimetric analyses of SBA-15 and SBA-TA samples. Inset figure: thermogram of crystalline TA.

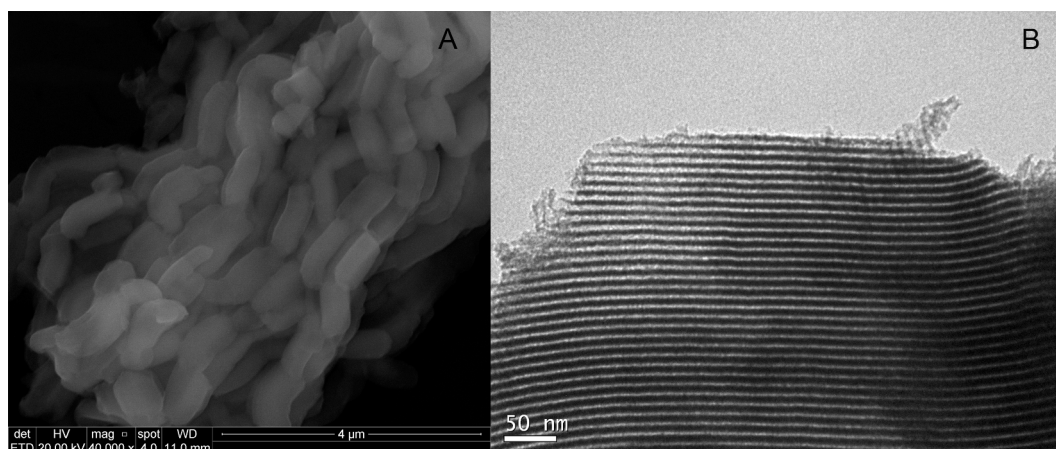


Fig. 1. Electron microscopy images of synthesized SBA-15: (A) SEM image at magnification of $40,000 \times$ and (B) TEM image at magnification of $300,000 \times$.

by condensation of silanol groups (Oliveira et al., 2020). These mass losses were observed in both samples, SBA-15 and SBA-TA. The organic decomposition of TA occurs in the same region, with emphasis on the region near 400 °C, where a greater loss of mass of the SBA-TA sample can be noted, compared to SBA-15. This is precisely the region of decomposition of TA, as observed in the inset Fig. 2.

Drug loading was calculated by subtracting the SBA-15 wt loss values from the SBA-TA in the range of 150 to 900 °C (Table S1). The loss of mass found by TGA (6.61%) corroborates the loading found by HPLC (6.70%) and correlates with the theoretical value (6.67%).

In order to investigate the textural properties of SBA-15 and the effect of drug loading in the porous structure for SBA-TA, N₂ adsorption-desorption isotherms were acquired, which are presented in Fig. 3A.

According to the isotherms, the samples showed a type IV pattern, which confirmed the presence of mesopores in the particles, with H1 type hysteresis indicating cylindrical pores with uniformity in the pore size (Sing et al., 1985). The SBA-15 sample presented a surface area of $796 \pm 5 \text{ m}^2 \text{ g}^{-1}$ and a pore volume of $1.083 \pm 0.008 \text{ cm}^3 \text{ g}^{-1}$. The SBA-TA formulation showed a reduction in the surface area and pore volume, presenting $600 \pm 2 \text{ m}^2 \text{ g}^{-1}$ and $0.862 \pm 0.002 \text{ cm}^3 \text{ g}^{-1}$, respectively, which is consistent with the filling process in the pores. When the BJH pore size distributions are analysed (Fig. 3B), SBA-15 has a maximum pore size distribution at 6.2 nm, while a slight shift of c.a. 0.2 nm in the pore size, to a lower diameter, is observed for the SBA-TA formulation. These results, along with the decrease in pore volume after the loading of the drug, indicate that TA is encapsulated in the mesopores and is evenly distributed along the pore walls as layers, not blocking their entry.

To characterize the solid state of the drug in SBA-TA, DSC analyses were performed (Fig. 4). The thermal profile of crystalline TA is characterized by the presence of an endothermic peak at 290 °C, related to the melting point of the drug (Tatke et al., 2019). As expected, SBA-15 did not show any peak in the same region. The same behaviour was observed for SBA-TA. The absence of the endothermic peak of crystalline TA could indicate that the drug was highly dispersed in the amorphous form throughout the SBA-15 surface. However, the curve related to the physical mixture also did not present the crystalline TA peak; therefore the absence of the crystalline TA signal in the SBA-TA curve might be due to a limitation of sensitivity in the technique.

For further investigation of the TA form in the SBA-TA, powder X-ray diffraction analysis was performed in the range of 2 θ from 3 to 60°, and the obtained diffractograms are presented in Fig. 5.

The diffractogram obtained for crystalline TA (Fig. 5) is in agreement with that reported by other authors (Pradhan et al., 2014). No diffraction pattern was observed in the SBA-15, only the amorphous halo of silica at 2 θ near 22°. On the other hand, for the SBA-TA formulation, besides the amorphous halo of silica, two main peaks of the TA diffraction pattern were still observed, indicating that, although the drug is highly dispersed in silica, a slight portion of TA can still be found in its crystalline form.

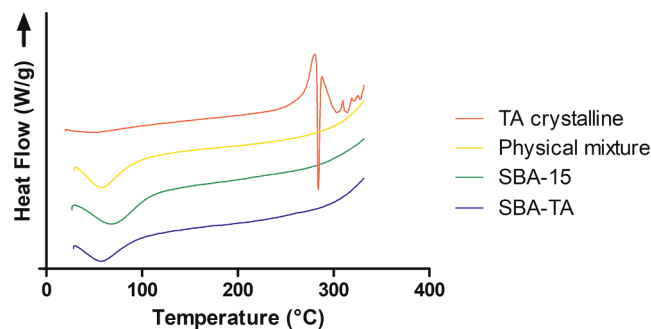


Fig. 4. DSC curves of crystalline TA, physical mixture, SBA-15 and SBA-TA samples.

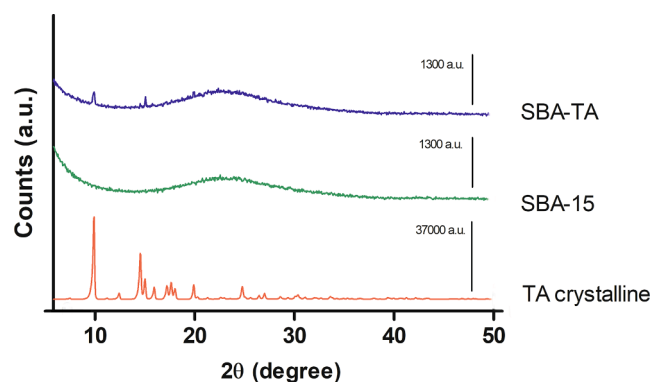


Fig. 5. X-ray diffractograms of SBA-15, SBA-TA and crystalline TA samples.

The dissolution/diffusion profiles of crystalline TA, TA in solution and the SBA-TA sample were evaluated using dialysis membranes. The cumulative percentage of dissolved crystalline TA, diffused TA from the ethanol solution and TA released from the SBA-TA sample were investigated *in vitro* over a period of 48 h. Each sample was analysed in triplicate, and the release curves are shown in Fig. 6. After withdrawal of the release medium, the TA in the samples was quantified by HPLC. In order to assay the drug in lower concentrations, the injection volume was adjusted to 50 μL , and the method remained linear ($r = 0.9953$) in the range of 0.1 to 2.5 $\mu\text{g mL}^{-1}$.

Complete release of the drug from the SBA-TA occurred after 12 h, with an identical release profile to that of the drug solubilized in ethanol being observed, unlike the crystalline TA (suspension), which took 48 h to be partially released (87%). By presenting the same release profile as the TA solution, SBA-TA demonstrates an improvement in the apparent aqueous solubility of the TA when encapsulated in the SBA-15 pores, taking into account that TA is in its solid form in the SBA-TA sample. A

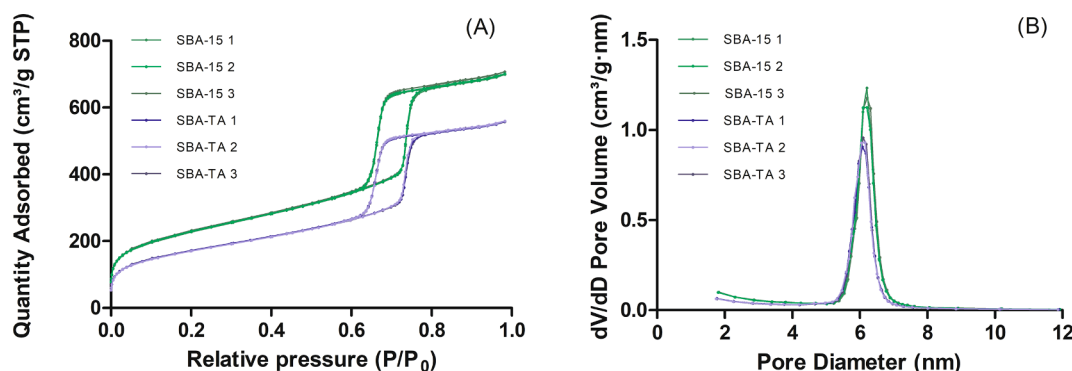


Fig. 3. Textural analysis ($n = 3$): (A) N₂ adsorption-desorption isotherms and (B) BJH pore size distribution.

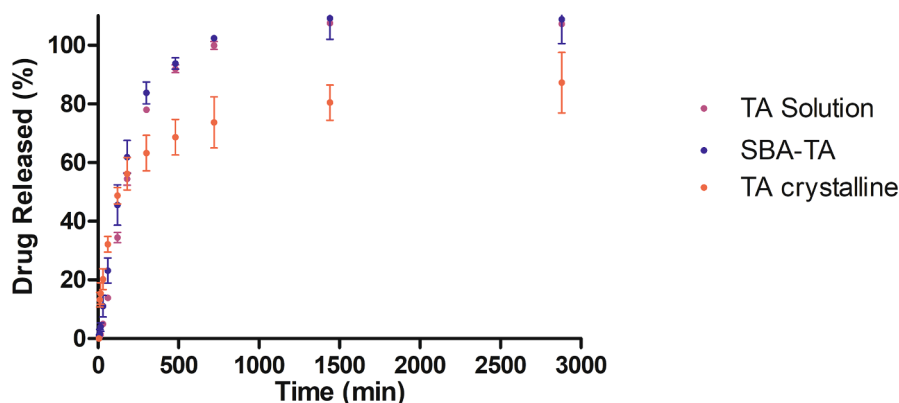


Fig. 6. TA *in vitro* dissolution profile ($n = 3$) from SBA-TA and free TA (ethanolic solution and crystalline). Absent standard deviation (SD) bars indicate SD < 1.

similar dissolution profile was reported by other authors (Patel et al., 2021) when incorporating riluzole in SBA-15, where the faster dispersion of the drug was attributed to the porous structure of the silica, which maintained the amorphous state of the drug, releasing 98% in 8 h.

3.2. Hydrogels and 3D-printed films of CMC containing SBA-TA

After characterization of the SBA-TA, this powder was incorporated into a CMC hydrogel in order to evaluate its use in the preparation of a 3D-printed solid dosage form. CMC was chosen as the polymer due to its property of forming hydrogels at low concentration and due to its wide application in 3D printing (Habib et al., 2018; Rahman et al., 2021). Glycerin was also added due to its hydrophilic groups, which could bind to mucin through hydrogen bonding, as well as improving swelling and, consequently, the flexibility of the film (Fernandes et al., 2018). Although TA has a melting point of about 263 °C, which could afford hot extrusion, the simple non-thermal extrusion of a semisolid technique was used in this approach, to allow its future application to drugs that could undergo degradation at temperatures above 40 °C.

The pH of the hydrogels were measured in order to investigate possible effects of irritation on the mucosa, since an alkaline or acidic pH can generate discomfort during administration (Sjöholm and Sandler, 2019). The pH for the hydrogels with and without the SBA-TA (HG-SBA-TA and HG, respectively) were neutral (7.7 ± 0.1), indicating that the film has suitable pharmaceutical characteristics for administration to the oral mucosa, although the pH of this region is slightly acidic (6.3–6.5) (Choi et al., 2013).

Regarding their rheological behavior, flow curves of both hydrogels (HG and HG-SBA-TA) showed a non-Newtonian flow with a shear-thinning behavior, i.e. a decrease in the apparent viscosity with the increase of shear rate (Fig. 7a), which is a desirable property to guarantee

extrudability of hydrogels during the printing process (Panraksa et al., 2021). Their rheological profiles were best fitted to the Herschel-Bulkley model ($R > 0.94$) according to the equation $\tau = \tau_0 + K\dot{\gamma}^n$ (τ = shear stress (Pa); τ_0 = yield stress (Pa); K = consistency index ($\text{Pa}\cdot\text{s}^2$); $\dot{\gamma}$ = shear rate (1/sec); n = flow behavior index). Herschel-Bulkley fluids are characterized by the need of a minimum force to start flowing, known as yield stress (Pa).

Yield stress values estimated by oscillatory stress sweep test showed that a higher force must be applied for HG to start to flow (2263 ± 47.31 Pa) in comparison with the HG-SBA-TA (2000 ± 63.73 Pa). Additionally, oscillation frequency test revealed a higher complex viscosity (η^*) of HG-SBA-TA (234.2 ± 6.8 Pa.s at 1 Hz) than HG (188.5 ± 8.4 Pa.s at 1 Hz). Therefore, a better flow during 3D printing extrusion might be achieved with the association of SBA-TA in the hydrogel despite its higher viscosity. In addition, both hydrogels presented G' higher than G'' , and a $\tan \delta$ (ratio of G''/G') below 1, indicating a predominant elastic behaviour, which is typical of gel-like formulations (Fig. 7b). Elasticity of hydrogels can indicate an easier recovery of their viscosity and shape retention after extrusion (Cheng et al., 2020).

The printed CMC films containing SBA-TA were homogeneous, colourless and had a smooth surface, as seen in Fig. 8. All films were easily removed from the glass surface on which they were printed, and were then placed on aluminium foil until dry. Good flexibility was observed when handling the films while still wet.

The weight of the films immediately after printing was 196 ± 14 mg; after drying at room temperature for 48 h, the weight reduced to 83 ± 20 mg. The same trend occurred when comparing their size, which visually reduced due to the partial loss of water content of the 3D printed film during the drying step. Its size after drying was 6.4 ± 0.4 mm, with 1.9 ± 0.4 mm in height.

The expected amount of drug per film was 84.9 ± 2.0 µg, which was

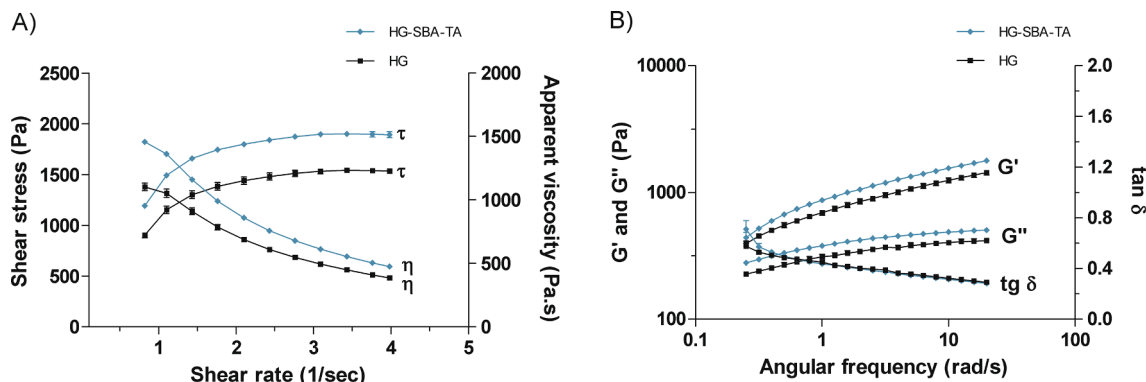


Fig. 7. Rheological data of HG and HG-SBA-TA. (A) Flow curves presented as shear stress (τ) and apparent viscosity (η) versus shear rate and (B) Oscillation frequency data of storage (G'), loss modulus (G'') and tangent delta ($\tan \delta$) versus angular frequency. Values are presented as mean $\pm$ SD ($n = 3$).

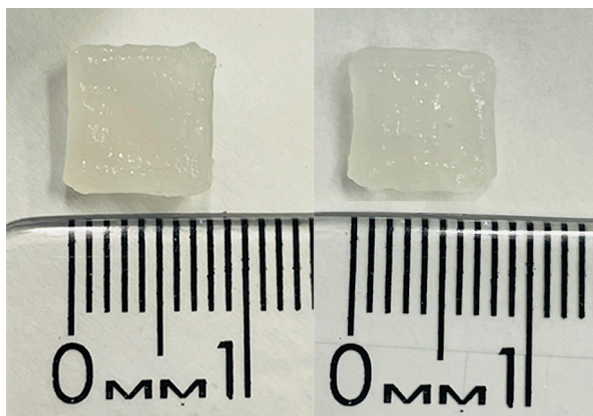


Fig. 8. Appearance and size of the 3D printed films of CMC containing SBA-TA.

calculated taking into account the film weight, the amount of SBA-TA incorporated into the hydrogel and the drug loading of SBA-TA. The experimental drug content in the films assayed by HPLC was $79.6 \pm 1.9 \mu\text{g}$ ($0.40 \pm 0.02 \text{ mg/g}$), representing an excellent and homogeneous recovery ($94 \pm 4\%$). Therefore, it is possible to suggest that the SBA-TA is evenly distributed in the hydrogel, affording 3D printed films with good reproducibility regarding drug content. Moreover, these data showed that there was no loss of drug during the incorporation of SBA-TA into the hydrogel and during the printing process.

Swelling studies for 3D printed films revealed swelling percentages of $249 \pm 14\%$ and $281 \pm 33\%$ for the films with and without SBA-TA after 30 min, respectively, with no significant difference between them ($p > 0.05$). The achieved swelling percentages indicate that the formulations are providing hydrogen-bond donors and acceptors in its structure that guarantees the strong hydrophilic character. This good swelling property of the films can positively influence the drug release kinetics and mucoadhesion of the films (Fernandes et al., 2018). After 30 min it was not possible to remove the films from the Petri dish glasses due to their fragmentation. This result is in agreement with the data from the disintegration test, where both films, with and without SBA-TA, had a disintegration time of c.a. 40 min, regardless of the presence of SBA-TA in the hydrogels.

Mechanical properties of the 3D printed films from plain CMC hydrogel and CMC hydrogels containing SBA-TA are presented in Table 1. The mechanical properties of 3D printed films can predict its capacity to ensure handling prior administration and transport, for example. The data demonstrated that films of CMC with SBA-TA needed a greater force to break when compared to the plain films, indicating their greater strength. The same trend was found from the stress test data. This improved strength of the film containing SBA-TA might be related to the higher viscosity presented by the original HG-SBA-TA. On the other hand, strain represents the percentage of deformation in relation to the original size of the film when the rupture occurs. This parameter is closely related to flexibility of the materials. In some cases, the more solid content of the formulation, the lower its flexibility (Sjöholm et al., 2020). The results found here showed that the incorporation of SBA-TA did not affect the flexibility of the films.

Table 1

Mechanical properties of 3D printed films with and without SBA-TA ($n = 4$, mean $\pm$ SD).

Sample	Force (N)	Distance (mm)	Stress (MPa)	Strain (%)
Plain 3D printed CMC film	$0.55 \pm 0.19^*$	73 ± 13	$0.244 \pm 0.083^*$	290 ± 56
3D printed CMC film loaded with SBA-TA	$1.00 \pm 0.10^*$	58 ± 5	$0.442 \pm 0.044^*$	230 ± 19

For each parameter, means with * are statistically different by *t* test ($p < 0.05$).

The dissolution profile obtained by the dialysis bag method is presented in Fig. 9 and indicated that half of the drug (55%) was released in the first 3 h. However, it took 48 h to reach 100% of drug released from the polymeric film, showing that the incorporation of SBA-TA in the polymeric films decreased the TA release rate compared with the data discussed in Fig. 6. This behaviour can be explained by the increase of the viscosity of the medium, due to the swelling and erosion of the polymer in <45 min, which decreases the diffusion rate of the drug (Sjöholm and Sandler, 2019).

To evaluate the mucoadhesive properties of the 3D printed films, an alternative mucin disc model was used, which was previously reported as suitable to discriminate the behaviour of different formulations (Chaves et al., 2018). It is known that mucoadhesive polymers generally contain hydrogen-bond donors and acceptors (carboxyl, hydroxyl or amino groups) and rapidly swell when hydrated. CMC is a semi-synthetic cellulose-derived polymer that, when in contact with mucin, is capable of forming hydrogen bonds. Also, the fast hydration of this polymer leads to a strong mucoadhesion property (Çelik et al., 2017). As shown in Table 2, the 3D printed films with SBA-TA showed higher work values (mN.mm) when compared to the powders (SBA-TA and SBA-15) and to the film prepared without the SBA-TA (plain film), representing higher adhesion values. Statistical analysis (*t*-test) showed that there was no significant difference between the mucoadhesion of the powders, but a difference could be seen between the films (with and without SBA-TA). Likewise, there was a difference in mucoadhesion when comparing the films with the powders. This indicates that the presence of SBA-TA improved the mucoadhesion of the polymeric film, promoting a longer interaction with mucin. Further studies should be carried out to elucidate the interactions that lead to this different behaviour.

Taking all our findings together we can propose the use of SBA-15 silica nanomaterial as an efficient nanocarrier to allow the incorporation of a hydrophobic drug, as triamcinolone acetonide, in printable hydrogels without using organic cosolvents and high content of surfactants. The printing process can be done at room temperature and the addition of the nanomaterial did not impair the printability of hydrogels, as their rheological properties were not highly changed. These findings open the door for further studies coupling the use of inorganic drug nanocarriers and SSE 3D printing in the development of personalised and on-demand pharmaceutical formulations.

4. Conclusions

SBA-15 mesoporous silica was an efficient drug carrier to increase the apparent solubility of the hydrophobic drug triamcinolone acetonide and its dissolution rate. The improvement of these drug properties could be explained by the drug encapsulation in the mesopores of SBA-15, which was confirmed by the decrease of its surface area. The drug-loaded nanomaterial could be easily dispersed in a hydrophilic ink

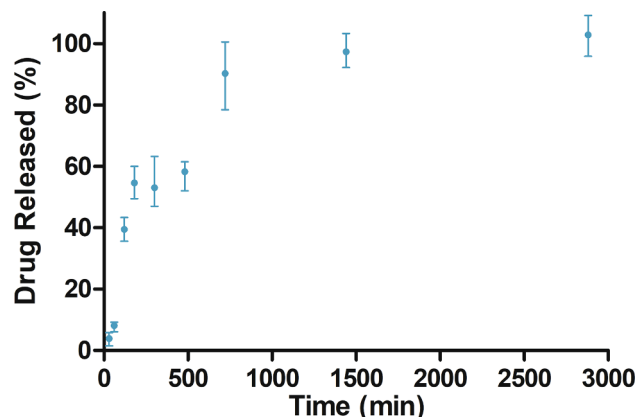


Fig. 9. *In vitro* dissolution profile of printed films of CMC containing SBA-TA.

Table 2

Work (mN.mm) required to displace the mucin disc from the 3D printed films (CMC plain film and CMC with SBA-TA film) and the powder samples (SBA-15 and SBA-TA).

	Sample	Work (mN.mm)
Powder	SBA-15	0.147 ± 0.15
	SBA-TA	3.105 ± 3.71
CMC 3D printed film	plain	151 ± 30*
	with SBA-TA	420 ± 54*

For the same pharmaceutical form, means with * are statistically different by *t* test (*p* < 0.05).

composed of CMC and SBA-TA, formulated as a hydrogel, which showed good printability by SSE. Its SSE 3D printing produced homogenous hydrophilic films showing complete drug release and improved *in vitro* mucoadhesion, highlighting them as a promising platform for the development of innovative oral mucosa delivery systems. This study represents the first report on the use of silica nanomaterials as drug carriers in the development of 3D printed dosage forms by SSE and opens the door to further studies on the use of this strategy to produce 3D printed customized nanomedicines.

CRedit authorship contribution statement

Laís Maltha Schmidt: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Juliana dos Santos:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Thayse Viana de Oliveira:** Investigation, Methodology, Writing – original draft. **Nadine Lysyk Funk:** Formal analysis, Investigation, Methodology, Writing – review & editing. **Cesar Liberato Petzhold:** Formal analysis, Investigation, Methodology, Writing – review & editing. **Edilson Valmir Benvenutti:** Formal analysis, Investigation, Methodology, Funding acquisition, Writing – original draft, Writing – review & editing. **Monique Deon:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Ruy Carlos Ruver Beck:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijpharm.2022.121750>.

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